

Figure 7.3 A map of the world using the default settings for D3's Mercator projection. You can see most of the Western Hemisphere and some of Europe and Africa, but the rest of the world is rendered out of sight.

Listing 7.4 Initial mapping function

```
d3.json("world.geojson", createMap);

function createMap(countries) {
  var aProjection = d3.geo.mercator();
  var geoPath = d3.geo.path().projection(aProjection);
  d3.select("svg").selectAll("path").data(countries.features)
    .enter()
    .append("path")
    .attr("d", geoPath)
    .attr("class", "countries");
};
```

Projection functions have many options that you'll see later.

d3.geo.path() defaults to albersUSA, which is a projection suitable only for maps of the United States.

d3.geo.path() takes properly formatted GeoJSON features and returns SVG drawing code for SVG paths.

Why do you only see part of the world in figure 7.3? Because the default settings of the Mercator projection show only part of the world in your SVG canvas. Each projection has a `.translate()` and `.scale()` that follow the syntax of the transform convention in SVG, but have different effects with different projections.

SCALE

You have to do some tricks to set the right scale for certain projects. For instance, with our Mercator projection if we divide the width of the available space by 2 and divide the quotient by `Math.pi`, then the result will be the proper scale to display the entire world in the available space. Figuring out the right scale for your map and your projection is typically done through experimenting with different values, but it's easier when you include zooming, as you'll see in section 7.2.2.